National University of Computer and Emerging Sciences, Lahore Campus

Course:	COAL	Course Code:	EE2003
Program:	BS(CS,DS)	Semester:	Fall 2022
Duration:	1 Hour	Total Marks:	30
Paper Date:	27-09-2022	Page(s):	5
Section:	All	Roll No.	
Exam:	Midterm-I	Your Section:	Solution.

Instruction/Notes: **This is an open notes/book exam. Sharing notes and calculators is NOT ALLOWED.** All the answers should be written in provided space on this paper. Rough sheets can be used but will not be collected and checked. In case of any ambiguity, make reasonable assumptions. Questions during exams are not allowed.

Question 1 [CLO 1] [3x5 = 15 Marks]: Short questions

i) For each of the instructions given below, **identify whether the instruction is valid or invalid:**

Instruction	Valid/Invalid
mov [Bp-si],2	Invalid
mov [bl], 2	Invalid
mov word[bx],2	valid
mov [bp+bx+si],2	Invalid
mov word[bx+ax],3	Invalid
mov word[num1], word[num2]	Invalid

ii) Given the following jump statements and their opcodes, **identify the type of jump (near or short) and the offset (value of IP) to which the jump will take place.** Show your complete working to get credit.

Offset of Opcode	Opcode	Type of Jump	Offset
011D	76E1 ;76 is opcode of jump	Short	0x0100
0100	E9FF1E ; E9 is opcode of jump	near	0x 2002

Show your working here:

...ing the instruction given below? Show your working to get full marks. ... the value

CS: 0x1E0A, DS: 0x1EED, SS: 0xFFEF, BP: 0x011F, SI: 0x0114, DI: 0x0112, BX: 0x0115

Instruction: Mov ax, [bp+si+15]

Value of AX after Execution of above instruction: 0009

Physical Memory Addresses	Memory Content
0x1F112	0A00
0x1E2E2	0500
0x00132	0900
0x00138	0700
0x1F118	0600

Show your working here:

```
  011F
  0114
+    F
------
  0242

  00242
  FFE20
-------
  00132
```

iv) Write assembly language code that checks if the contents of two registers **AX** and **BX** are equal and jumps LabelXYZ (if contents are equal). You are NOT ALLOWED to use ***cmp, sub*** and ***sbb*** instructions. No credit will given if your code exceeds two lines.

```
xor ax, bx
jz labelxyz
```

v) For the code snippet given below, write the byte sized data value ... execution of the program. Briefly explain the working of this program.

Value of Lab2 after execution of program: -1

Write values of following flags after instruction "cmp dl, [bx+lab1]" (highlighted in code) for first iteration only.

SF = 1 OF= 0 CF= 1 ZF= 0

Code Snippet:	Show your working here (for flag values):
see below	~~1000~~

```
[org 0x100]
        jmp start

lab1:   db -3, -1, 4
lab2:   db 0

start:  mov bx, 1
        mov dl, [lab1]

loop1:  cmp dl, [bx+lab1]
        jae C1

        mov dl, [bx+lab1]

C1:     add bx, 1
        cmp bx, 3
        jne loop1

        mov [lab2], dl

        mov ax, 0x4c00
        int 0x21
```

What is above code doing? Describe in one line only.

Max unsigned

Scanned with CamScanner

Q.2 Solution.

```
    mov si, 0
    mov di, 0

l1:   mov bx, 0
      mov al, [set1 + si]
      cmp al, [set2 + bx]
      jge l2
      inc bx, 1
      cmp bx, 5
      jne l1
      mov cl, 0
      mov cl, [set1+si]
      mov byte [difference + di], cl
      inc di, 1

l2:   inc si, 1
      cmp si, 7
      jne l1
```